PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Maximum Marks: 4 Marks

Technical Architecture

The solution is implemented in ServiceNow Incident Management. Client Scripts provide client-side form behavior, while UI Policies control field visibility, mandatory status, and read-only behavior without changing the server-side data model.
Architecture Flow:
Flow: User opens/updates an Incident → UI Policy evaluates conditions → Client Script executes form logic when applicable → form fields are updated → ServiceNow processes and stores the Incident record.

Table-1: Components & Technologies

Table-2: Application Characteristics

Implementation Details

1. Client Script – Incident

Purpose: Implement client-side behavior on the Incident form, such as validating field values, displaying messages, and dynamically setting field values.

```
// Example: Incident - onChange Client Script
function onChange(control, oldValue, newValue, isLoading, isTemplate) {
if (isLoading || newValue == '') {
return;
}

// Example business rule:
// If Priority is critical, require a business justification message.
if (newValue == '1') {
g_form.showFieldMsg(
'priority',
'Critical priority selected. Confirm that the incident impact and urgency are justified.',
'warning'
);
}
}
```

2. UI Policy – Incident

Purpose: Dynamically control Incident form fields using declarative conditions. For example, when an Incident is resolved, resolution-related fields can be made mandatory.

```
UI Policy:
Table: Incident [incident]
Condition: State is Resolved

UI Policy Actions:
- Resolution code: Mandatory = True
- Resolution notes: Mandatory = True
- Resolution code / notes: Visible = True
- Fields controlled by the policy can be made Read-only after closure if required.

Reverse if false:
- Restore the normal field behavior when the condition is no longer true.
```

Expected Outcome

The Incident form responds dynamically to user actions. Client Scripts handle client-side validation and interaction, while UI Policies manage declarative field behavior. The implementation improves consistency of Incident data entry and reduces unnecessary manual checks by agents.

References

ServiceNow documentation for Client Scripts, UI Policies, GlideForm (g_form), Incident Management, and Access Control Lists.
Reference template supplied for Project Design Phase-II – Technology Stack (Architecture & Stack).

Date	29 September 2026
Team ID	SWTID-2026-7872
Project Name	ServiceNow Incident – Client Script and UI Policy Implementation

ServiceNow Incident Form	UI Policy Rules	Client Script Logic	Incident Record / Form State	Server-side ServiceNow Platform

S.No	Component	Description	Technology
1	User Interface	ServiceNow Incident form used by agents to create and update incidents.	ServiceNow UI / Incident form
2	Application Logic-1	Client-side logic to validate or react to Incident field changes.	ServiceNow Client Script / JavaScript
3	Application Logic-2	Declarative form behavior such as mandatory, visible, or read-only fields based on conditions.	ServiceNow UI Policy
4	Application Logic-3	Optional form messaging and dynamic field updates for Incident users.	GlideForm (g_form) / JavaScript
5	Database	Stores Incident records and field values.	ServiceNow platform database
6	Cloud Database	Managed platform data persistence for the ServiceNow application.	ServiceNow cloud platform
7	File Storage	Attachments associated with Incident records, when required.	ServiceNow Attachments
8	External API-1	Not required for the basic Client Script/UI Policy implementation.	Not applicable
9	External API-2	Not required for the basic Client Script/UI Policy implementation.	Not applicable
10	Machine Learning Model	Not required for the basic Incident form customization.	Not applicable
11	Infrastructure (Server / Cloud)	ServiceNow hosts the application, platform services, database, and runtime.	ServiceNow Cloud / SaaS

S.No	Component	Description	Technology
1	Open-Source Frameworks	No external open-source framework is required. The implementation uses native ServiceNow configuration and JavaScript.	ServiceNow Client Script / JavaScript
2	Security	Access is controlled through ServiceNow roles, ACLs, application permissions, and platform security. Client-side controls are not treated as server-side security.	ServiceNow ACLs / Roles / Platform Security
3	Scalable Architecture	Form behavior is separated into declarative UI Policies and Client Scripts, allowing changes without modifying the core Incident application.	ServiceNow platform architecture
4	Availability	The application runs on the ServiceNow hosted platform, using the platform's managed availability and operational infrastructure.	ServiceNow Cloud

S.No	Component	Description	Technology
5	Performance	Use lightweight client-side logic, condition-based UI Policies, and avoid unnecessary server calls from Client Scripts.	JavaScript / GlideForm / UI Policies